

JBT Financial Midterm Report

David T. Wilkey
Developer
JBT Financial
Knoxville, TN

Bhumir Patel
Developer
JBT Financial
Knoxville, TN

Joey Syracuse
Developer
JBT Financial
Knoxville, TN

Abstract—Planning investments based on market trends and research is a time consuming process that many individuals are unwilling/unable to pursue. Even when extensive market research is performed, many investors are still left with a sense of uncertainty surrounding their financial choices. By leveraging linear regression, random forest, and k-nearest neighbor techniques, JBT Financial has developed machine learning tools which analyze vast amounts of financial data quickly and efficiently to provide the user with valuable insights to help guide their investment decisions. This paper provides an overview of the machine learning pipelines used throughout the development process and discusses preliminary findings based on established baseline models.

I. INTRODUCTION

The rise in popularity of mobile brokerage platforms has made stock trading more accessible for both large and small investors. JBT Financial has developed new tools to address many challenges faced by stock traders.

II. LINEAR REGRESSION MODEL DEVELOPMENT

A. Data Exploration

B. Baseline Linear Regression Model

III. RANDOM FOREST MODEL DEVELOPMENT

A. Data Exploration

B. Baseline Random Forest Model

IV. K-NEAREST NEIGHBOR MODEL DEVELOPMENT

A. Data Exploration

B. Baseline K-Nearest Neighbor Model

V. PLANNED MODEL IMPROVEMENTS

VI. CONCLUSION

REFERENCES

Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2]. Refer simply to the reference number, as in [3]—do not use “Ref. [3]” or “reference [3]” except at the beginning of a sentence: “Reference [3] was the first . . .”

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be

cited as “unpublished” [4]. Papers that have been accepted for publication should be cited as “in press” [5]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

REFERENCES

- [1] G. Eason, B. Noble, and I. N. Sneddon, “On certain integrals of Lipschitz-Hankel type involving products of Bessel functions,” *Phil. Trans. Roy. Soc. London*, vol. A247, pp. 529–551, April 1955.
- [2] J. Clerk Maxwell, *A Treatise on Electricity and Magnetism*, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68–73.
- [3] I. S. Jacobs and C. P. Bean, “Fine particles, thin films and exchange anisotropy,” in *Magnetism*, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271–350.
- [4] K. Elissa, “Title of paper if known,” unpublished.
- [5] R. Nicole, “Title of paper with only first word capitalized,” *J. Name Stand. Abbrev.*, in press.
- [6] Y. Yorozu, M. Hirano, K. Oka, and Y. Tagawa, “Electron spectroscopy studies on magneto-optical media and plastic substrate interface,” *IEEE Transl. J. Magn. Japan*, vol. 2, pp. 740–741, August 1987 [Digests 9th Annual Conf. Magnetism Japan, p. 301, 1982].
- [7] M. Young, *The Technical Writer’s Handbook*. Mill Valley, CA: University Science, 1989.

IEEE conference templates contain guidance text for composing and formatting conference papers. Please ensure that all template text is removed from your conference paper prior to submission to the conference. Failure to remove the template text from your paper may result in your paper not being published.